

BIONIC EYE MODEL TO PROVIDE VISION OR RESTORE SIGHT FOR BLINDNESS USING VISION TRANSFORMER

A PROJECT REPORT

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KOVILACHERY, KUMBAKONAM



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BONAFIDECERTIFICATE

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DECLARATION

We declare that the work entitled “**BIONIC EYE MODEL TO PROVIDE VISION OR RESTORE SIGHT FOR BLINDNESS USING VISION TRANSFORMER**” is submitted in partial fulfillment of the requirement for the award of degree in **BACHELOR OF INFORMATION TECHNOLOGY**, ANNAUNIVERSITY,CHENNAI, is a record of our own work carried out by us during the academic year 2025-2026 under the supervision and guidance of **Mrs.B.BHARATHI,M.E.**, Assistant Professor of Information Technology, Anna College of Engineering & Technology. The extent and source of information derived from existing literature have been indicated through the dissertation at the appropriate places. The matter embodied in this work is original and has been submitted for the award of any degree or diploma, either in this or any university.

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ABSTRACT

Visual implants are intended to produce an artificial vision leading to some levels of functional vision restoration. It uses 60 microelectrodes implanted in the retina and can improve the quality of life of visually impaired people by making them experience light even if they were in the dark for many years. Due to the limited number of microelectrodes of existing visual system stimulator, the artificial vision they permit has very low resolution. Many researchers have worked on improving the artificial vision created with low resolution implants by using image processing and machine vision algorithms. Users' express dissatisfaction with the Retinal Prosthesis System due to the low resolution of phosphene images, highlighting the critical need for focused research to enhance visual clarity and improve overall user satisfaction. This project proposes a simulation of the artificial vision in which the information synthesized by the system to the visually impaired user using a visual implant generated low resolution phosphene image. By employing Vision Transformer (ViT), the method extracts valuable information about individuals surrounding the visually impaired user, such as their count, familiarity, gender, estimated ages, facial emotions, surrounding objects and approximate distances. This data, derived from camera frames on the user's glasses, is utilized to generate signals fed into a visual stimulator, presenting a promising approach to enrich the visual experience for individuals with visual impairments.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
OCR	Optical character recognition
AIML	Artificial intelligence markup Language
RPN	Region Proposal Network
LBP	Local Binary Patterns
CNN	Convolutional Neural Network
MiDAS	Multiple Depth Estimation Accuracy With Single Network
DL	Deep Learning

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Vision is the complex, multipart process that gives sense of sight. Vision itself doesn't happen in eyes. Instead, it's the sum of eyes, retinas, optic nerves and brain working together to process reflected light from the world around us. Maintaining vision can make a big difference in well-being and quality of life. Also called low vision, visual impairment is defined as loss of eyesight that cannot be corrected with glasses, medicine, or surgery. For people who are visually impaired, everyday tasks such as reading, shopping, recognizing faces, and even crossing the street may be difficult.

1.1.1. Three Types of Vision Impairments

The types of vision impairments are low visual acuity, blindness, and legal blindness (which varies for each country):

- Low visual acuity, also known as moderate visual impairment, is a visual acuity between 20/70 and 20/400 with your best corrected vision, or a visual field of no more than 20 degrees.
- Blindness is a visual acuity of 20/400 or worse with your best corrected vision, or a visual field of no more than 10 degrees.
- Legal blindness in the United States is a visual acuity of 20/200 or worse with your best corrected vision or a visual field of no more than 20 degrees.

Being blind or visually impaired doesn't mean you automatically lose the independence of getting to and from places whenever you please. Several techniques and methods can help people get around safely regardless of their amount of vision.

1.2 PROBLEM STATEMENT

The biggest challenge for a blind person, especially the one with the complete loss of vision, is to navigate around places. Obviously, blind people roam easily around their house without any help because they know the position of everything in the house. People living with and visiting blind people must make sure not to move things around without informing or asking the blind person. Commercial places can be made easily accessible for the blinds with tactile tiles. But unfortunately, this is not done in most of the places. This creates a big problem for blind people who might want to visit the place. Firstly, there has traditionally been an overwhelming emphasis on ICT tasks such as reading or using a computer, rather than helping those with vision loss better interpret real-world scenes. Portability and convenience remain a second major problem and, even though many such issues have been significantly mitigated by the emergence of the smartphone over the past ten years, blind people have long been crying out for solutions that are more seamlessly integrated and wearable. The most valuable thing for a disabled person is gaining independence. A blind person can lead an independent life with some specifically designed adaptive things for them. There are lots of adaptive equipment that can enable a blind person to live their life independently but they are not easily available in the local shops or markets. Assistive technology to help those who are blind or severely sight impaired has, for several decades, been dogged by a number of critical issues. Over the past few years, there has been increasing interest in AI powered smart glasses. Artificial intelligence & machine learning can be deployed to create products for enhanced productivity among the differently-abled.

1.3 MACHINE LEARNING

1.3.1 RPN

Step 1: Input Image

- The input image is fed into the neural network, typically a convolutional neural network (CNN), to extract features through convolutional layers.

Step 2: Anchor Boxes

- A set of anchor boxes (also known as anchor proposals or priors) is defined at various scales and aspect ratios. These anchor boxes serve as potential bounding box proposals for objects in the image.

Step 3: Convolutional Layer

- The features extracted from the input image are passed through a convolutional layer. This layer helps to capture hierarchical features and representations of objects in different spatial scales.

Step 4: Intermediate Layer for Classification

- The output of the convolutional layer is then passed through an intermediate layer for classification. This layer predicts two scores for each anchor box: one for the probability of the anchor containing an object (foreground) and another for the probability of it being background.

Step 5: Intermediate Layer for Regression

- Another intermediate layer is used for regression to refine the coordinates of the anchor boxes. This layer predicts adjustments (delta values) to the dimensions (width and height) and location (x, y) of the anchor boxes to better fit the actual objects in the image.

Step 6: Anchor Box Scores

- The predicted scores for each anchor box (foreground and background) are used to calculate the objectness score. Objectness score is obtained by applying the softmax function to the predicted foreground and background scores.

Step 7: Non-Maximum Suppression (NMS)

- Non-Maximum Suppression is applied to filter out redundant bounding box proposals.
- Bounding boxes with lower objectness scores are suppressed if they overlap significantly with boxes having higher scores.

Step 8: Top-N Proposals

- The top-N proposals with the highest objectness scores are selected as the final bounding box proposals.
- These proposals are used as inputs for subsequent stages of the object detection pipeline, such as RoI (Region of Interest) pooling or RoI alignment.

1.4 OBJECTIVE

The aim of the project is to enhance the artificial vision experience for visually impaired individuals through the integration of advanced technologies, such as Vision Transformer and image processing algorithms. The goal is to provide a more comprehensive and clear artificial vision system that improves the quality of life, mobility, and independence for visually impaired users.

Other objectives of the project may include:

- **To Implement Vision Transformer Technology:** Integrate Vision Transformer (ViT) technology for improved artificial vision clarity.
- **To Extract Comprehensive Information:** Develop real-time algorithms for extracting people count, familiarity, gender, age, emotions, objects, and distances.
- **To Enable Real-time Image Processing:** Implement efficient real-time image processing algorithms for live video analysis.
- **To Enhance Accessibility with Audio Output:** Integrate audio output with text-to-speech conversion for effective communication to visually impaired users.
- **To Validate with Simulated Prosthetic Vision:** Validate the system with a simulated

prosthetic vision, accommodating 150 microelectrodes for compatibility.

- To Handle Versatile Tasks: Develop algorithms capable of handling tasks including face recognition, gender prediction, age estimation, emotion prediction, and object detection.
- To Assess User Satisfaction: Evaluate user satisfaction through testing, feedback, and comparison with existing visual implant solutions.
- To Explore Integration with Visual Implants: Explore and demonstrate compatibility with existing and future visual implant technologies.

CHAPTER 2

LITERATURE SURVEY

2.1 Title: Enhanced Artificial Vision for Visually Impaired.

Author: Hossein Mahvash Mohammadi, Mohammad Hadi

Using visual implants, capture, process images, identify objects, and stimulate the brain, providing enhanced artificial vision for visually impaired individuals which Enhanced Independence and Improved quality of life. Through a sophisticated process, these implants capture visual information from the environment, process it to identify objects, and then stimulate the brain to create artificial vision. This technology essentially bypasses the damaged or non-functional parts of the visual system, allowing users to perceive their surroundings in a way that was previously impossible.

2.2 Title: Voice Navigation Based Guiding Device for Visually Impaired People.

Author: Chitra, Sumath.

CNN (convolution neural network algorithm) Findings: Find out the experiments was conducted with few objects such as chair, bicycle, person and a water bottle. The proposed method can detect the obstacles very effectively. Which is Low False Positive Rate and Effective Obstacle Detection.

2.3 Title: Guidance System for Visually Impaired People.

Author: Xu Zhang, Lin Liu, Luiz Xiao.

OCR (optical character recognition) AIML (artificial intelligence markup Language).The system is aimed to provide voice-over assistants for blind people to do tasks like understanding surroundings, looking for an object, recognizing the face of a person with emotion, and reading etc.

CHAPTER 3

PROBLEM DEFINITION

3.1 PROBLEM STATEMENT

The biggest challenge for a blind person, especially the one with the complete loss of vision, is to navigate around places. Obviously, blind people roam easily around their house without any help because they know the position of everything in the house. People living with and visiting blind people must make sure not to move things around without informing or asking the blind person. Commercial places can be made easily accessible for the blinds with tactile tiles. But unfortunately, this is not done in most of the places. This creates a big problem for blind people who might want to visit the place. Firstly, there has traditionally been an overwhelming emphasis on ICT tasks such as reading or using a computer, rather than helping those with vision loss better interpret real-world scenes. Portability and convenience remain a second major problem and, even though many such issues have been significantly mitigated by the emergence of the smartphone over the past ten years, blind people have long been crying out for solutions that are more seamlessly integrated and wearable. The most valuable thing for a disabled person is gaining independence. A blind person can lead an independent life with some specifically designed adaptive things for them. There are lots of adaptive equipment that can enable a blind person to live their life independently, but they are not easily available in the local shops or markets. Assistive technology to help those who are blind or severely sight impaired has, for several decades, been dogged by several critical issues. Over the past few years, there has been increasing interest in AI powered smart glasses. Artificial intelligence & machine learning can be deployed to create products for enhanced productivity among the differently abled.

3.2 EXISTING SYSTEM

A Kinect-Based Navigation System The navigation system described here requires a standard Kinect sensor, a battery, and a laptop/processor, all of which are carried in a backpack or shoulder bag by the user. The Kinect is powered by a sealed lead acid 12 V, 7 A battery shown in Figure 3, the output of which is fed through a DC-to-DC converter to ensure a stable 12 V supply. The capacity of the battery is enough to power both Kinect and a portable CPU for 3 hours.

Disadvantage

- Existing system appear to be due to the infrared sensor as it starts giving blind spots (no depth values) under strong sunlight.
- Identifying objects and face at reasonable distances and speeds
- Most portable designs seem to resort to mechanical components, thus causing a conflict between their bulkiness and the subtlety of the induced perceptions.
- Very Expensive
- False Prediction
- Time taken to predict.

3.3 PROPOSED SYSTEM

In the initial phase of Artificial Vision Build and Train, the project focuses on assembling a diverse dataset, gathering images relevant to face recognition, gender prediction, age estimation, and emotion prediction. Following dataset collection, meticulous pre-processing steps, including grayscale conversion, resizing, noise filtering, and binarization, ensure standardized input conditions for optimal model training. Advanced segmentation techniques like the Region Proposal Network (RPN) are then applied to identify and extract facial regions, providing a foundation for subsequent analyses. Feature extraction, utilizing methods like Local Binary

Patterns (LBP), captures distinctive features crucial for nuanced tasks. The final step involves implementing a Convolutional Neural Network (CNN) for unified classification tasks, consolidating face recognition, gender prediction, age estimation, and emotion prediction into a cohesive framework.

Vision Prediction System using Vision Transformer

In the Vision Prediction System using Vision Transformer, real-time video is captured through a device-mounted camera, serving as input for subsequent analysis by the Vision Transformer. This technology predicts whether individuals in the user's vicinity are known or unknown, extracting features and enhancing spatial awareness. In the presence of unknown individuals, the system predicts gender, age, and facial expression. The MiDaS algorithm estimates distances, and object detection identifies and locates objects, contributing to heightened situational awareness. To convey predictions, the system utilizes audio output with text-to-speech conversion, ensuring accessible feedback for visually impaired users.

Advantages

- Enhances artificial vision for better clarity.
- Promotes greater independence for visually impaired users.
- Integrates real-time image processing for swift and dynamic analyses of
- live video feeds.
- Audio output ensures effective information delivery.
- Provides timely information for better situational awareness.

CHAPTER 4

SYSTEM STUDY

4.1 FEASIBILITY STUDY

This project is feasible provided given unlimited resources and infinite time. Unfortunately, the development of a computer-based system is more likely to be plagued by resource scarcity and stringent schedules. It is both necessary and prudent to evaluate the feasibility of a project at earliest possible time. Wastage of manpower and financial resources and untold professional embarrassment can be avoided if an ill-conceived system is recognized early in the development phase. So a detailed study was carried out to check the workability of the proposed system. Feasibility study is a test of system proposal regarding its workability, impact on the organization, ability to meet user needs and effective use of resources. Thus, when an application is proposed, it is normally goes through a feasibility study before it is approved for development.

Feasibility and risk analysis is related in many ways. If project risk is great, the feasibility of producing quality is reduced. Thus, during feasibility analysis for this project, following three primary areas for interest was considered very carefully.

There are several types of feasibility. Three key considerations involved in the feasibility analysis are,

- **Technical Feasibility**
- **Operational Feasibility**
- **Economic Feasibility**

Technical Feasibility

This study is carried out to check the technical feasibility; this is the technical requirements of the system. Any system developed must not have a importance to

consider the monetary factors also it might happen that developing a particular system may be technically possible. This will lead to high demands being placed on the client.

The developed system must have modest requirements, as only minimal or null changes are required for implementing this system.

In this technical feasibility the following issues are taken into considerations. Once the technical feasibility is established. For evaluating this, economic feasibility of the proposed system is carried out.

Operational Feasibility

The proposed system normally solves the problem and takes advantages of the opportunities identified during scope definitions; it satisfies the requirements identified in the requirement analysis phase of system development. Since the statistical figures are stored in a certain format in the computer, reduce the manual work and enhance the standard of presentation also.

Non-operational feasibility assesses the extent to which the required software performs a series of steps to solve business problems and user requirements. This measures how well your company will be able to solve problems and take advantage of opportunities that are presented during the project.

Economic Feasibility

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

In economic feasibility, cost benefits analysis is done in which expected cost and benefits are evaluated. Economic analysis is used for evaluating the effectiveness of the proposed system. The developed system is economical when compared to the existing system job done manual system. So, the proposed system is so fast that planning can be made easily.

CHAPTER 5

SYSTEM SPECIFICATION

5.1 HARDWARE REQUIREMENTS

- Processors : Intel® Core™ i5 processor
- RAM : 16 GB
- Disk space : 320 GB
- Operating systems : Windows 10

5.2 SOFTWARE REQUIREMENTS

- Front End : Python 3.7.4(64-bit) or (32-bit)
- IDE : Flask 1.1
- Back End : MySQL 5
- Server : WampServer 2i
- Packages : Pandas, Sikit Learn, NumPy, matplotlib, seaborn
- Map : Google Map API

SOFTWARE DESCRIPTION

5.3. PYTHON3.7.4

Python is a general-purpose interpreted, interactive, object-oriented, and high-level programming language. It was created by Guido van Rossum during 1985-1990. Like Perl, Python source code is also available under the GNU General Public License (GPL). This tutorial gives enough understanding on Python programming language.



Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable. Python is a MUST for students and working professionals to become a great Software Engineer specially when they are working in Web Development Domain.

Python is currently the most widely used multi-purpose, high-level programming language. Python allows programming in Object-Oriented and Procedural paradigms. Programmers must type relatively less and indentation requirement of the language, makes them readable all the time. Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram,

Dropbox, Uber... etc. The biggest strength of Python is huge collection of standard libraries which can be used for the following:

- Machine Learning
- GUI Applications (like Kivy, Tkinter, PyQt etc.)
- Web frameworks like Django (used by YouTube, Instagram, Dropbox)
- Image processing (like OpenCV, Pillow)
- Web scraping (like Scrapy, BeautifulSoup, Selenium)
- Test frameworks
- Multimedia
- Scientific computing
- Text processing and many more.

5.3.1 Pandas

pandas is a fast, powerful, flexible and easy to use open source data analysis and manipulation tool, built on top of the Python programming language. pandas is a Python package that provides fast, flexible, and expressive data structures designed to make working with "relational" or "labeled" data both easy and intuitive. real world data analysis in Python.



Pandas is mainly used for data analysis and associated manipulation of tabular data in Data frames. Pandas allows importing data from various file formats such as comma-separated values, JSON, Parquet, SQL database tables or queries, and Microsoft Excel. Pandas allows various data manipulation operations such as merging, reshaping, selecting, as well as data cleaning, and data wrangling features. The development of pandas introduced into Python many comparable features of working.

with Data frames that were established in the R programming language. The panda's library is built upon another library NumPy, which is oriented to efficiently working with arrays instead of the features of working on Data frames.

5.3.2 NumPy

NumPy, which stands for Numerical Python, is a library consisting of multidimensional array objects and a collection of routines for processing those arrays. Using NumPy, mathematical and logical operations on arrays can be performed.



NumPy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays.

5.3.3 Matplotlib

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. Matplotlib makes easy things easy and hard things possible.



Matplotlib is a plotting library for the Python programming language and its numerical mathematics extension NumPy. It provides an object-oriented API for embedding plots into applications using general-purpose GUI toolkits like Tkinter, wxPython, Qt, or GTK.

5.3.4 Seaborn

Seaborn is a library for making statistical graphics in Python. It builds on top of matplotlib and integrates closely with pandas data structures. Visualization is the central part of Seaborn which helps in exploration and understanding of data.

Seaborn offers the following functionalities:

- Dataset oriented API to determine the relationship between variables.
- Automatic estimation and plotting of linear regression plots.
- It supports high-level abstractions for multi-plot grids.
- Visualizing univariate and bivariate distribution.

5.3.5 Scikit Learn

scikit-learn is a Python module for machine learning built on top of SciPy and is distributed under the 3-Clause BSD license.



Scikit-learn (formerly scikits. learn and also known as sklearn) is a free software machine learning library for the Python programming language. It features various classification, regression and clustering algorithms including support-vector machines, random forests, gradient boosting, k- means and DBSCAN, and is designed to interoperate with the Python numerical and scientific libraries NumPy and SciPy.

5.3.6 MYSQL

MySQL tutorial provides basic and advanced concepts of MySQL. Our MySQL tutorial is designed for beginners and professionals. MySQL is a relational database management system based on the Structured Query Language, which is the popular language for accessing and managing the records in the database. MySQL is open-source and free software under the GNU license. It is supported by Oracle Company. MySQL database that provides for how to manage database and to manipulate data with the help of various SQL queries. These queries are: insert records, update records, delete records, select records, create tables, drop tables, etc.



MySQL is currently the most popular database management system software used for managing the relational database. It is open-source database software, which is supported by Oracle Company. It is fast, scalable, and easy to use database management system in comparison with Microsoft SQL Server and Oracle Database. It is commonly used in conjunction with PHP scripts for creating powerful and dynamic server-side or web-based enterprise applications. It is developed, marketed, and supported by MySQL AB, a Swedish company, and written in C programming language and C++ programming language. The official pronunciation of MySQL is not the My Sequel; it is My Ess Que Ell. However, you can pronounce it in your way. Many small and big companies use MySQL. MySQL supports many Operating Systems like Windows, Linux, MacOS, etc. with C, C++, and Java languages.

5.3.7 WAMP SERVER

WampServer is a Windows web development environment. It allows you to create web applications with Apache2, PHP and a MySQL database. Alongside, PhpMyAdmin allows you to manage easily your database.



WampServer is a reliable web development software program that lets you create web apps with MYSQL database and PHP Apache2. With an intuitive interface, the application features numerous functionalities and makes it the preferred choice of developers from around the world. The software is free to use and doesn't require a payment or subscription.

5.3.8 FRONT END:

DESIGNBOOTSTRAP

Bootstrap is a free and open-source tool collection for creating responsive websites and web applications. It is the most popular HTML, CSS, and JavaScript framework for developing responsive, mobile-first websites.



It solves many problems which we had once, one of which is the cross-browser compatibility issue. Nowadays, the websites are perfect for all the browsers (IE, Firefox, and Chrome) and for all sizes of screens (Desktop, Tablets, Phablets, and Phones). **Easy to use:** Anybody with just basic knowledge of HTML and CSS can start using Bootstrap.

Responsive features: Bootstrap's responsive CSS adjusts to phones, tablets, and desktops.

Mobile-first approach: In Bootstrap, mobile-first styles are part of the core framework.

Browser compatibility: Bootstrap 4 is compatible with all modern browsers (Chrome, Firefox, Internet Explorer 10+, Edge, Safari, and Opera)

5.3.9 FLASK

Flask is a web framework. This means flask provides you with tools, libraries and technologies that allow you to build a web application. This web application can be some web pages, a blog, a wiki or go as big as a web-based calendar application or a commercial website.

Using an IDE

As good as dedicated program editors can be for your programming productivity, their utility pales into insignificance when compared to Integrated Developing Environments (IDEs), which offer many additional features such as in-editor debugging and program testing, as well as function descriptions and much more.



Flask is often referred to as a micro framework. It aims to keep the core of an application simple yet extensible. Flask does not have built-in abstraction layer for database handling, nor does it have formed a validation support. Instead, Flask supports the extensions to add such functionality to the application. Although Flask is rather young compared to most Python frameworks, it holds a great promise and has already gained popularity among Python web developers. Let's take a closer look into Flask, so-called "micro" framework for Python. Flask was designed to be easy to use and extend. The idea behind Flask is to build a solid foundation for web applications of different complexity. From then on you are free to plug in any extensions you think you need. Also, you are free to build your own modules. Flask

is great for all kinds of projects. It's especially good for prototyping. Flask is part of the categories of the micro-framework. Micro-framework is normally framework with little to no dependencies to external libraries. This has pros and cons. Pros would be that the framework is light, there are little dependency to update and watch for security bugs, cons is that sometimes you will have to do more work by yourself or increase yourself the list of dependencies by adding plugins.

CHAPTER 6

SYSTEM DESIGN

6.1 SYSTEM ARCHITECTURE

An architecture diagram is a graphic representation that shows how various software system components will be physically implemented. It displays the overall architecture of the software system along with the relationships, constraints, and divisions among its many components.

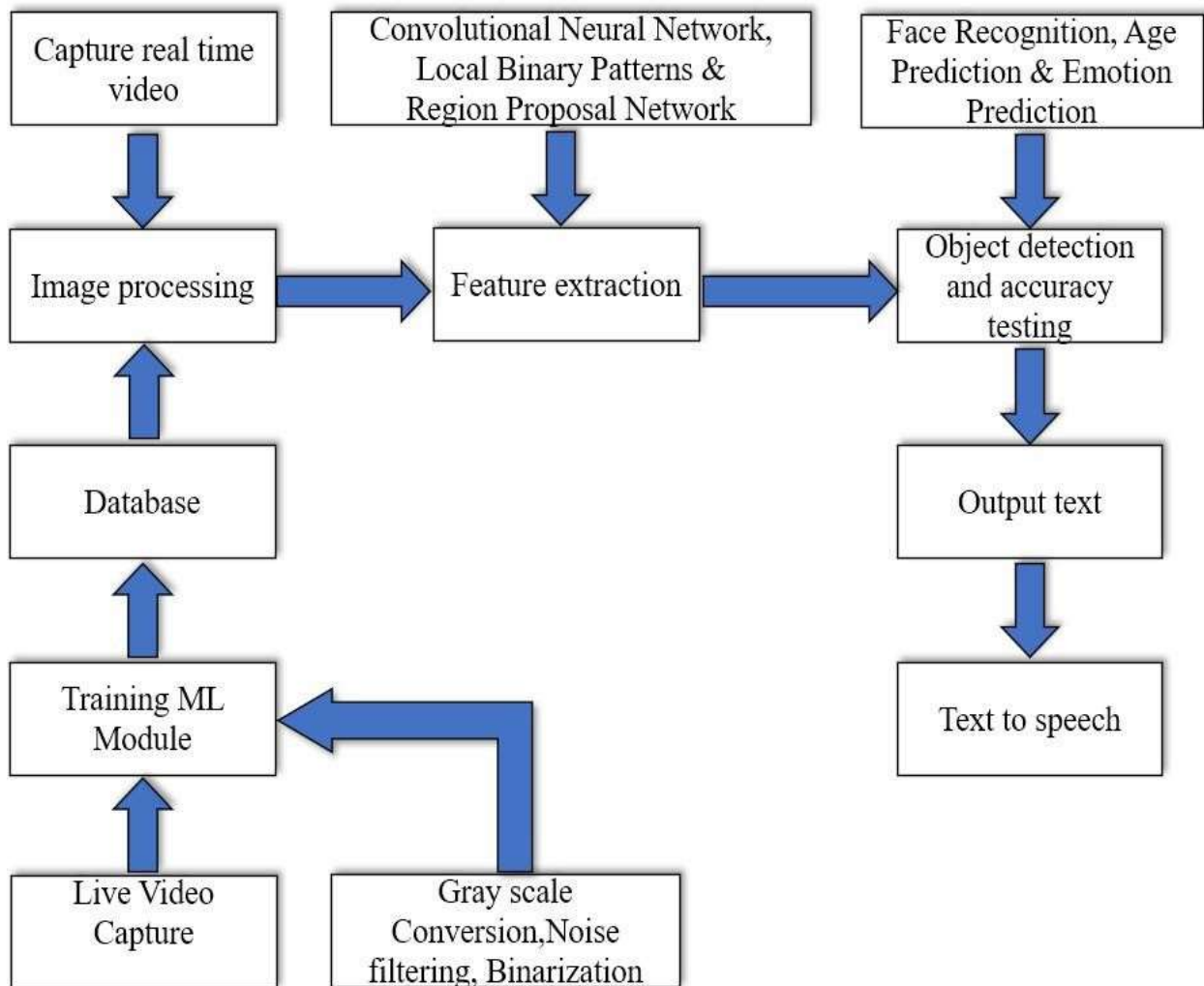


Figure 6.1.1 System Architecture

DATAFLOW DIAGRAM

A data flow diagram (DFD) is a graphical representation of data through an information system. It is a useful tool for analyzing, modelling, and documenting a system's processes and the flow of data between those processes. DFDs can help you understand the relationships between different components and identify potential bottlenecks or inefficiencies.

LEVEL0

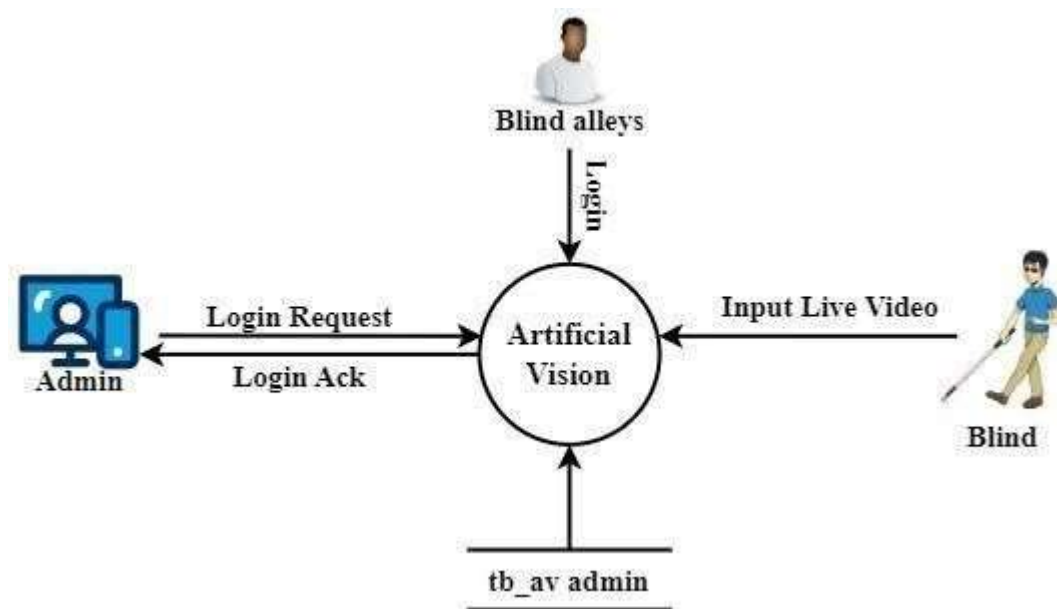


Figure 6.2.1 Dataflow Diagram

LEVEL 1

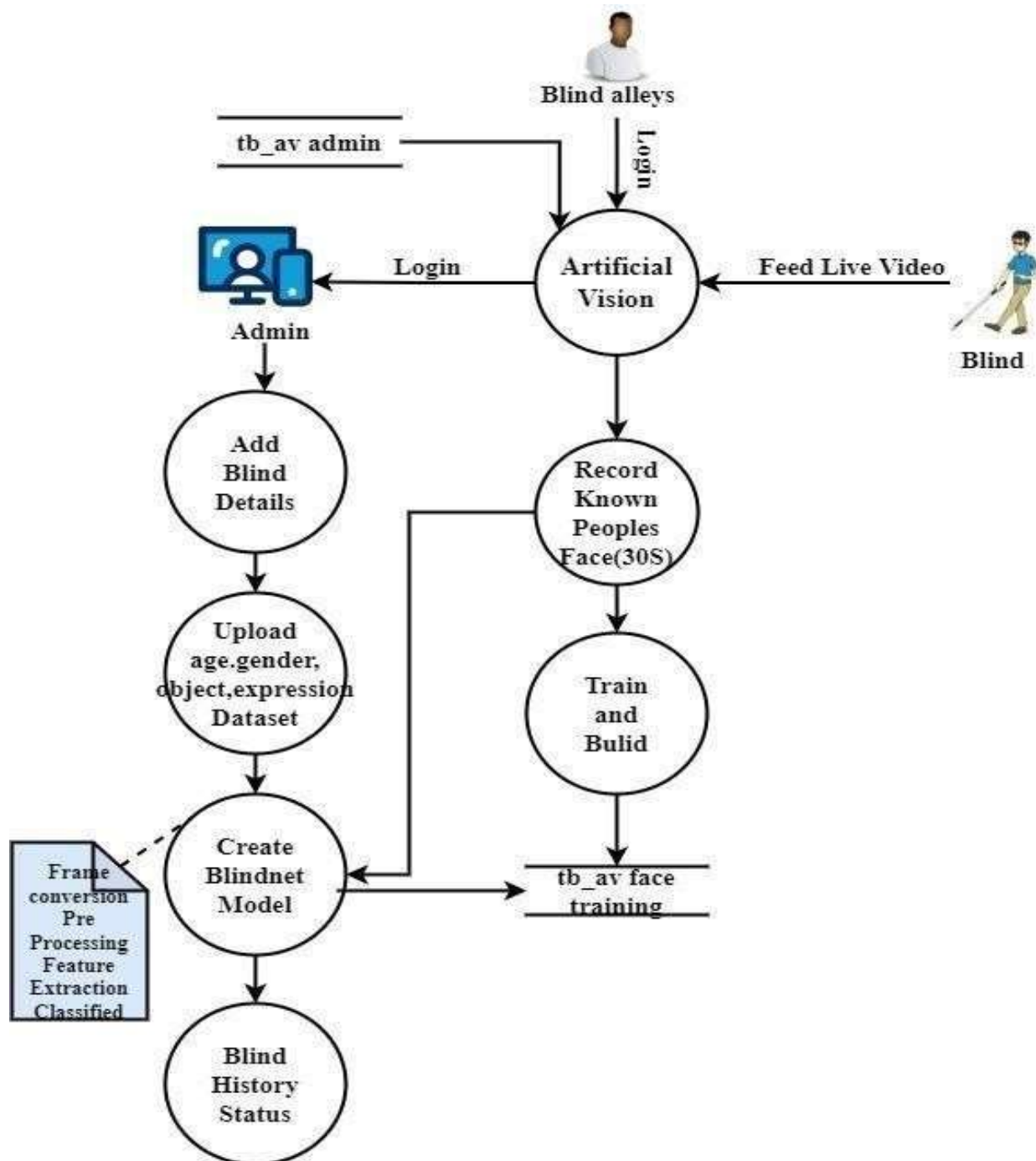


Figure 6.2.2 Dataflow Diagram

LEVEL 2

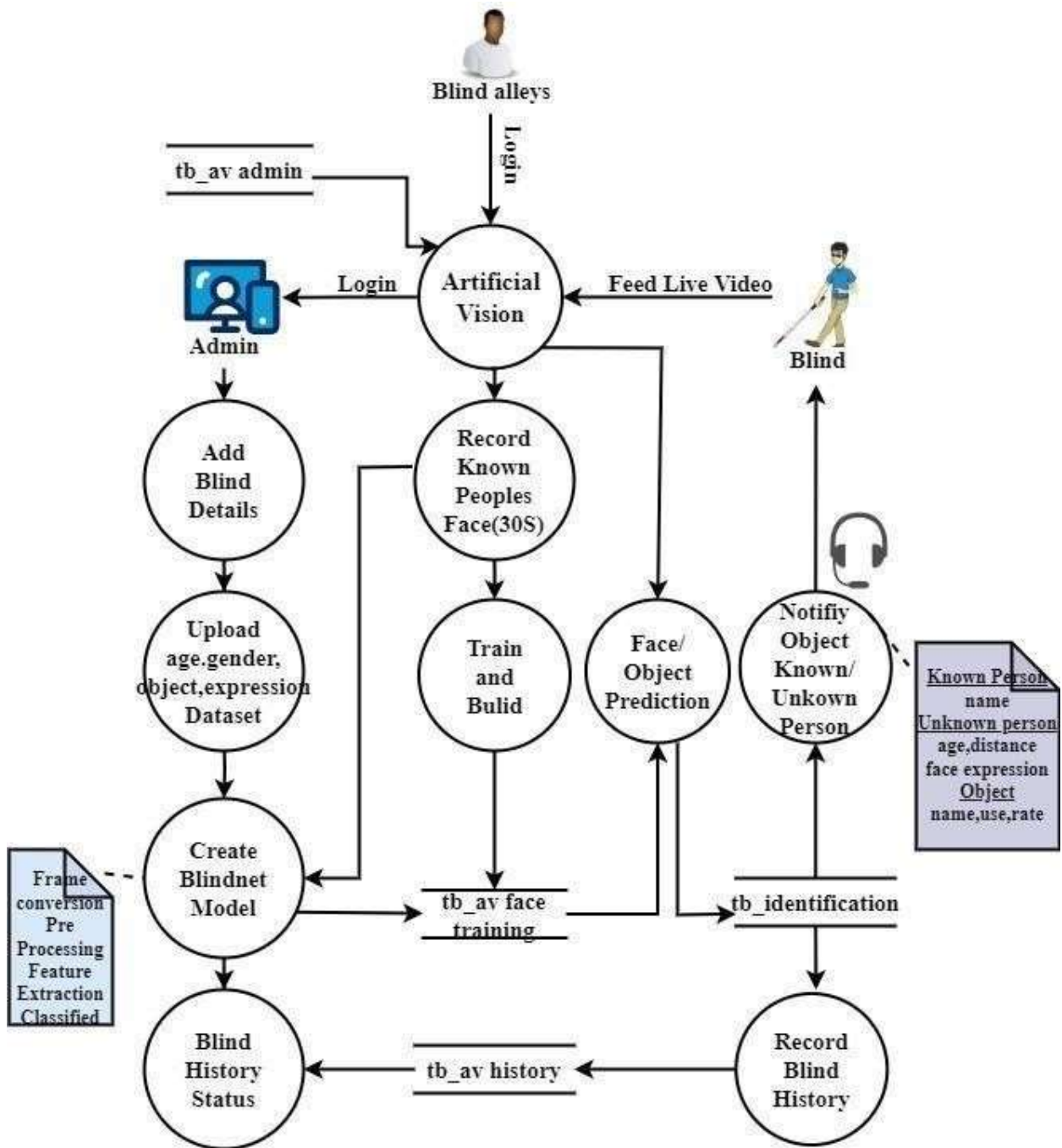


Figure 6.2.3 Dataflow Diagram

6.3. UML DIAGRAM

A UML diagram is a diagram based on the UML (Unified Modeling Language) with the purpose of visually representing a system along with its main actors, roles, actions, artifacts or classes, in order to better understand, alter, maintain, or document information about the system.

6.3.1. USE CASE DIAGRAM

A use case diagram is a type of UML diagram that illustrates the interactions between actors and system or application. It represents the functionality of the system and the specific tasks or actions that can be performed by the actors. Use case diagrams are useful for identifying the requirements of a system, defining the scope of a project, and communicating the functionality of a system to stakeholders.

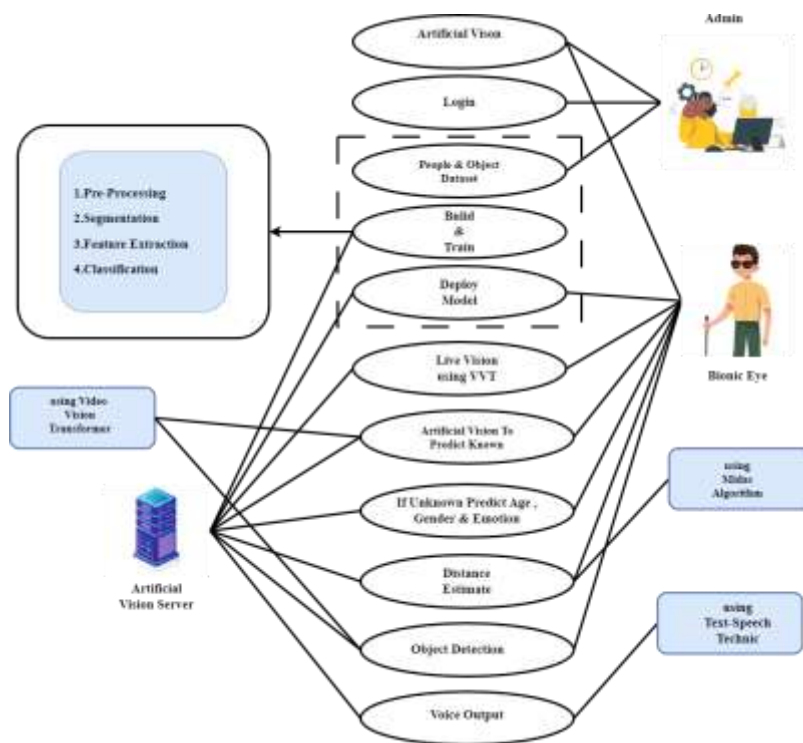


Figure 6.3.1 Use case diagram

6.3.2. CLASS DIAGRAM

A class diagram in the unified modelling language (UML) is a type of static structure diagram that describes the structure of a system by showing a system's classes, their attributes operations (or methods), and the relationships among objects. The class diagram is the main building block of object-oriented modeling of the systematic of the applications, and for detailed modeling translating the models into programming code. Class diagram can also be used for data modeling. The classes in a class diagram represent boy the main elements, interactions in the applications, and the classes to be programmed.

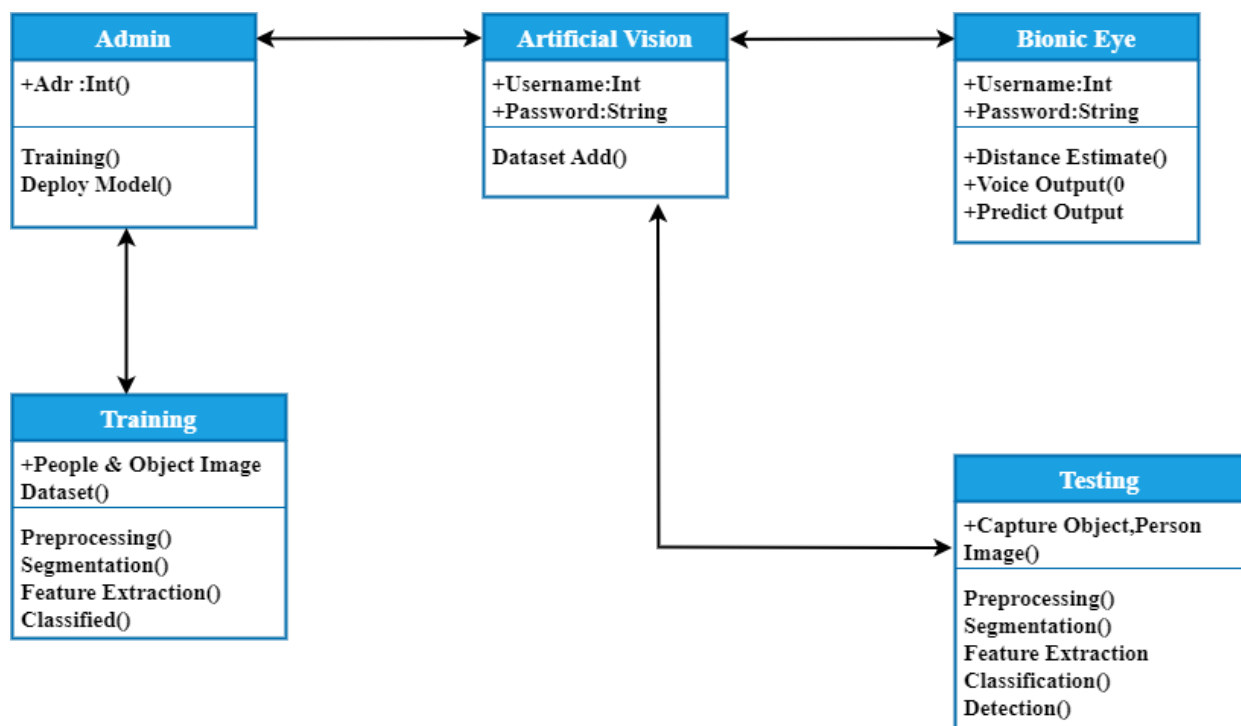


Figure 6.3.2 Class diagram

6.3.3. SEQUENCE DIAGRAM

A sequence diagram is an interaction diagram that shows how objects operate with one another and in what order. It is a construct of a message sequence chart. A sequence diagram shows object interactions arranged in time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario. Sequence diagrams are typically associated with use case realizations in the logical view of the system under development. Sequence diagrams are sometimes called event diagrams or event scenarios.

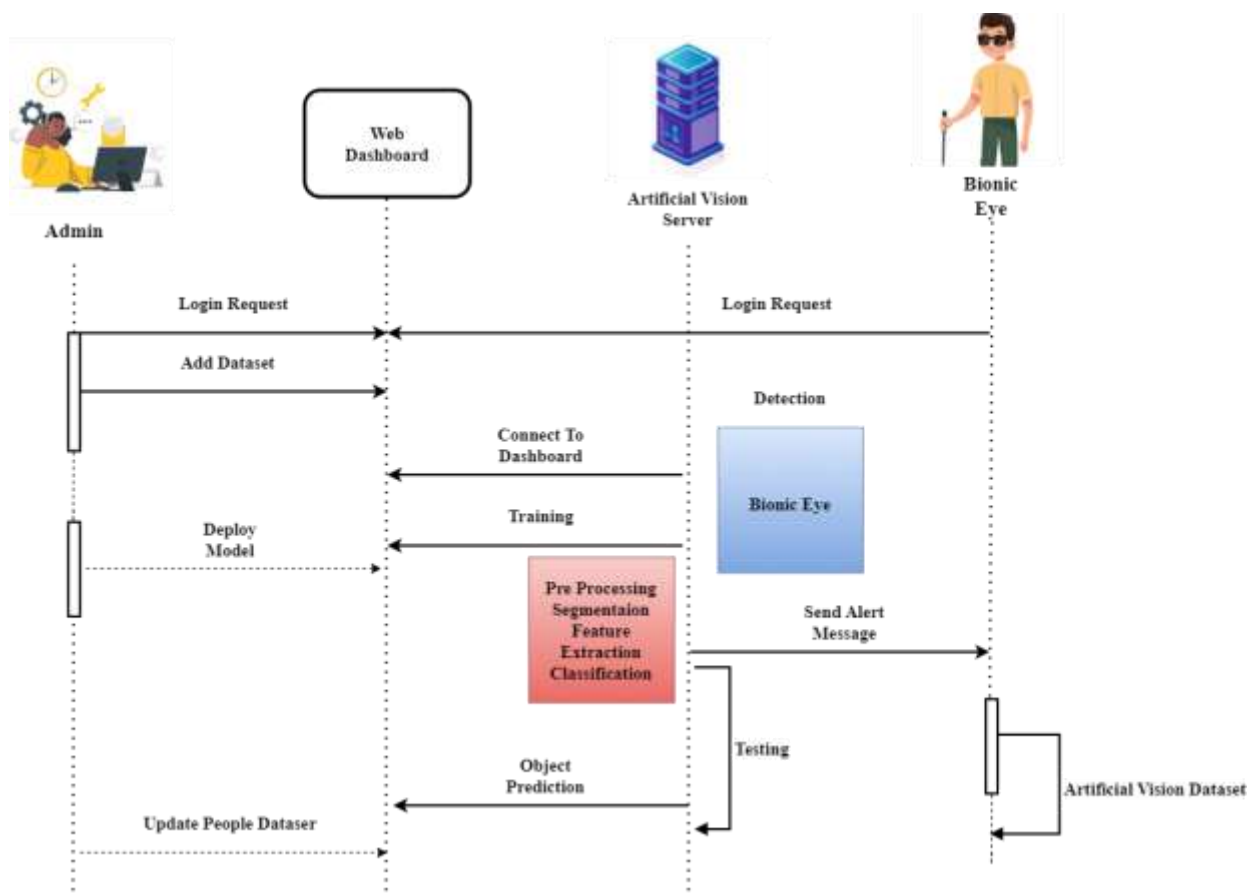


Figure 6.3.4 sequence diagram

6.3.4. ACTIVITY DIAGRAM

Activity diagrams show the workflow from a start point to the finish point detailing the many decision paths that exist in the progression of events contained in the activity. They may be used to detail situations where parallel processing may occur in the execution of some activities.

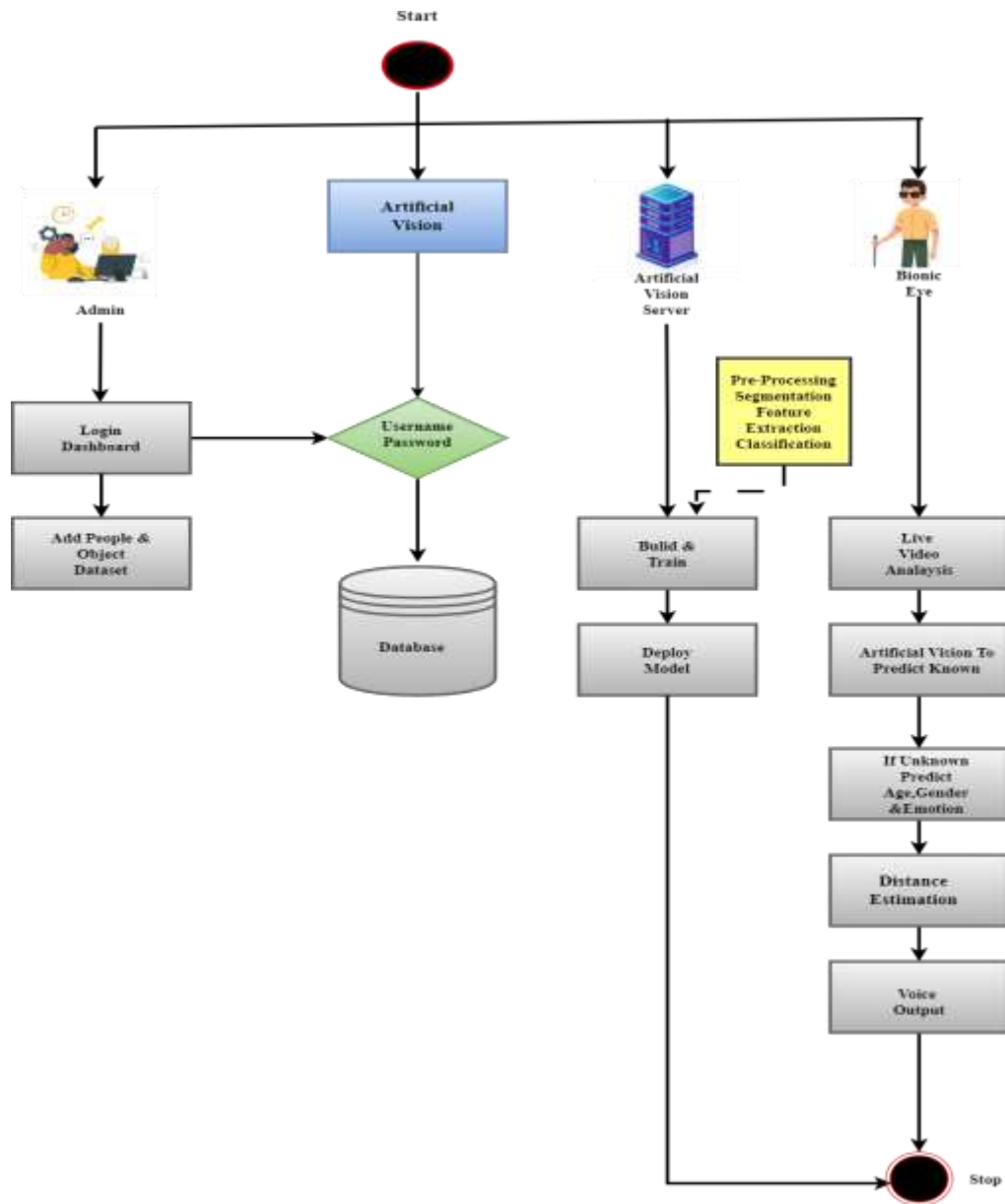


Figure 6.3.4 Activity diagram

CHAPTER 7

SYSTEM IMPLEMENTATION

SYSTEM DESCRIPTION

System implementation is the stage in the project where the theoretical design is turned into a working system. The most critical stage is achieving a successfully system and in giving confidence on the new system for the user that it will work efficiently and effectively.

1. Admin Training Module:

- The system starts with the admin accessing the training module.
- Admin has the option to train the system on both persons and objects.
- For person training, the admin provides labeled images for face recognition, gender prediction, age estimation, and emotion prediction.
- For object training, the admin provides labeled images to teach the system to recognize and classify different objects.

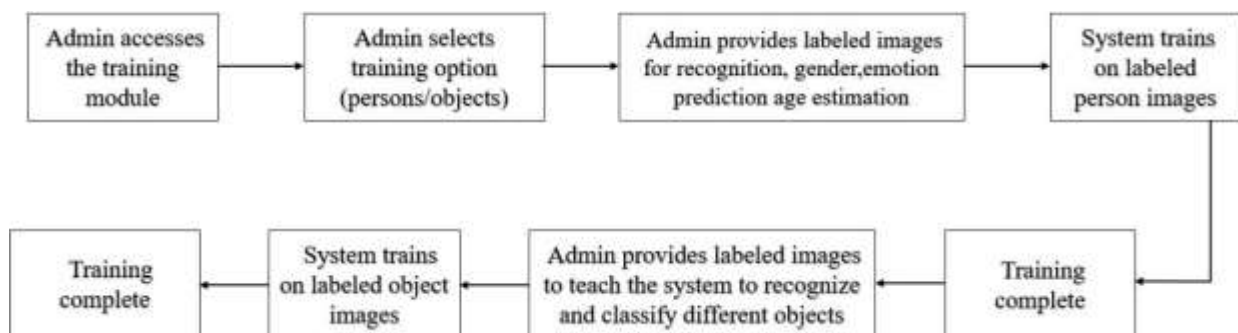


Figure 7.1.1 Admin Training Module diagram

2. Dataset Management:

- The labeled datasets for persons and objects are managed and stored in the database.
- The system ensures the datasets are well-organized and ready for model training.



Figure 7.2.1 Dataset Management diagram

3. Model Training:

- The system initiates model training using the provided datasets.
- The training process involves building and adjusting a machine learning model, such as a Convolutional Neural Network (CNN), for accurate recognition and prediction.

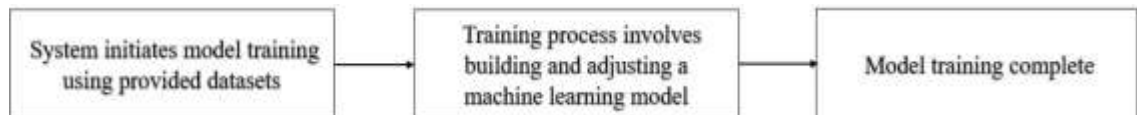


Figure 7.3.1 Model Training diagram

4. Admin Feedback and Iteration:

- The admin receives feedback on the training process, allowing adjustments and improvements.
- The iterative nature of this process ensures that the model becomes more accurate over time.

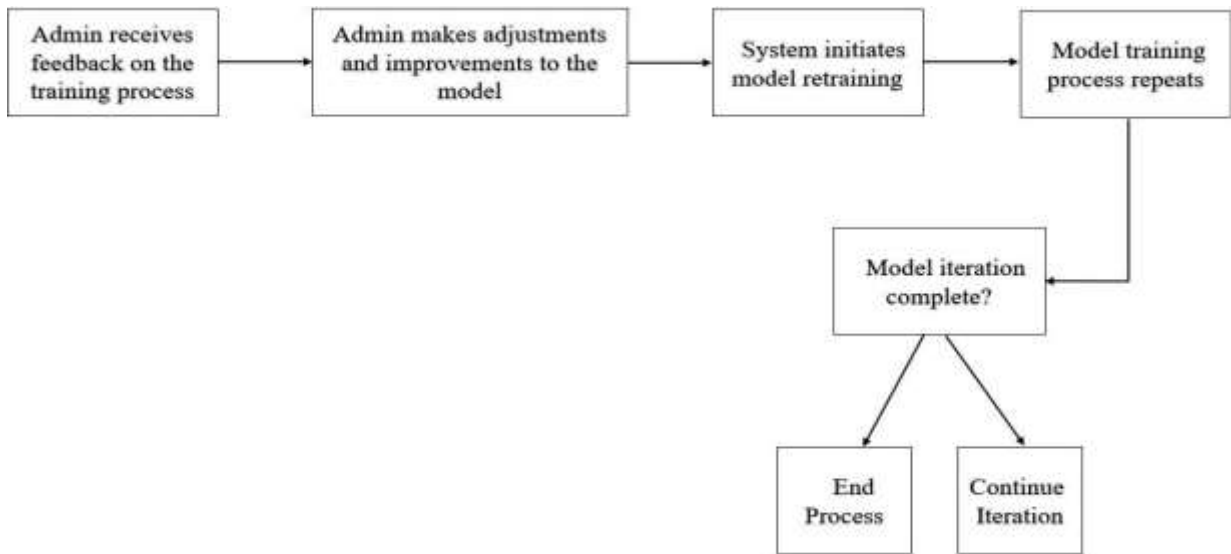


Figure 7.4.1 Admin Feedback and Iteration diagram

5. Blind User Live Video Module:

- The Blind user engages with the live video module using a camera mounted on their device.
- The live video feed serves as input for real-time analysis.

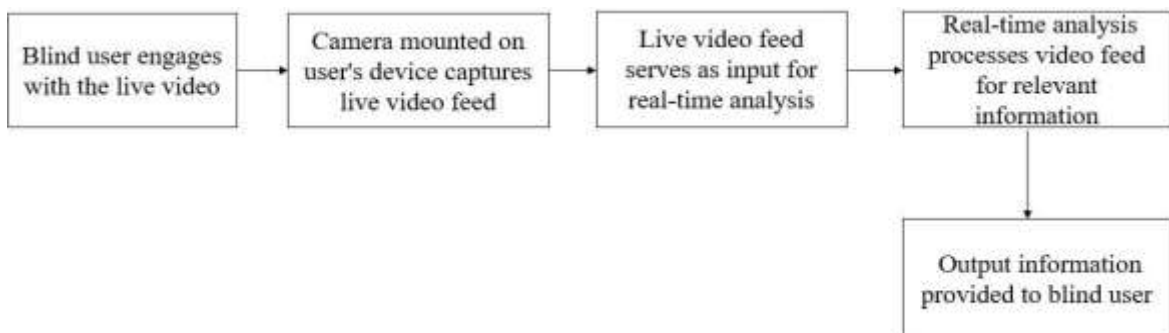


Figure 7.5.1 Blind User Live Video Module diagram

6. Vision Prediction System:

- The Vision Prediction System, incorporating trained models, analyzes the live video feed.
- It distinguishes between known and unknown individuals, predicts

gender, estimates ages, recognizes facial emotions, and identifies objects in the surroundings.

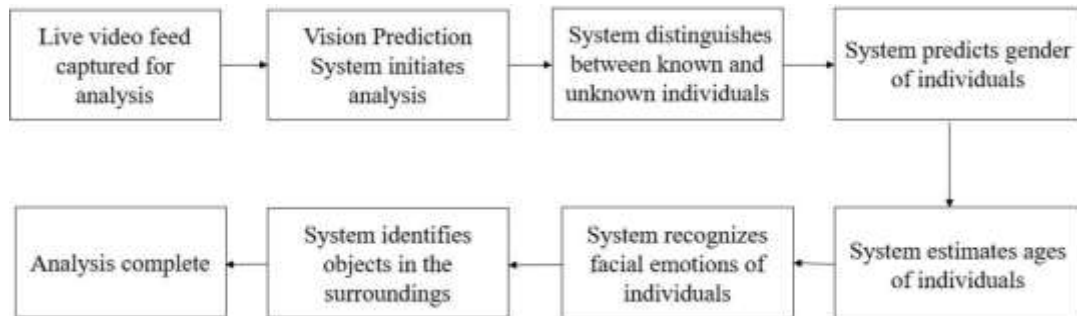


Figure 7.6.1 Vision Prediction System diagram

7. Spatial Awareness and Object Detection:

- The system estimates distances using algorithms like MiDaS for spatial awareness.
- Object detection identifies and locates objects within the live video feed.

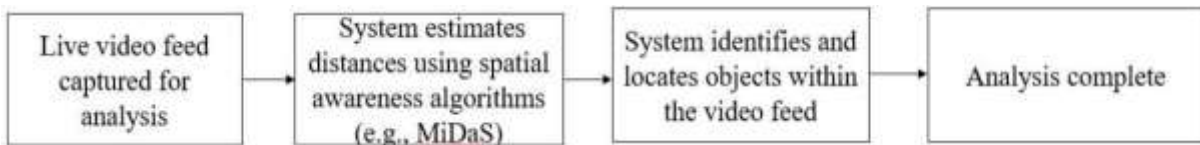


Figure 7.7.1 Spatial Awareness and Object Detection diagram

8. Audio Output for Blind User:

- Predictions and information are converted into audio cues using text-to-speech capabilities.
- The Blind user receives auditory feedback, including information about persons, objects, and the spatial lay

CHAPTER 8

SYSTEM TESTING

8.1 System Testing

Software testing is a critical phase in the software development life cycle that involves the systematic evaluation of a software application to ensure its quality, functionality, and performance.

Objective:

- Identify and fix defects or errors in the software.
- Enhance the quality, reliability, and performance of the software.

Types of Testing:

- **Manual Testing:** Testers manually execute test cases without using any automation tools. It involves the tester's observation and analysis of the application's behavior.
- **Automated Testing:** Testers use automation tools to execute test cases. Automation is beneficial for repetitive testing tasks, regression testing, and performance testing.

Testing Techniques:

- **Black Box Testing:** Testing the software without knowledge of its internal code or logic. Focus is on inputs and outputs.
- **White Box Testing:** Examining the internal code, logic, and structure of the software. Requires knowledge of the internal workings.

Testing Life Cycle:

- **Test Planning:** Defining the overall testing approach and strategy.
- **Test Design:** Creating test cases based on requirements and specifications.
- **Test Execution:** Running the test cases and capturing results.
- **Test Closure:** Summarizing the testing process and outcomes.

Types of Software Testing

1. Functional Testing:

- **Unit Testing**

Unit testing involves testing individual units or components in isolation to ensure they function as intended. It is typically performed by developers during the coding phase to catch and fix bugs early.

- **Integration Testing**

Integration testing checks the interaction between integrated components or systems. It verifies that the combined units work together seamlessly and identifies any issues that may arise from their integration.

- **System Testing**

System testing evaluates the entire software system to ensure it meets specified requirements. It involves testing the software in its entirety to validate its functionality, performance, and reliability.

- **Acceptance Testing**

Acceptance testing is performed to ensure that the software meets acceptance criteria and is ready for deployment. It involves validating whether the software satisfies user requirements and business expectations.

2. Non-Functional Testing

- **Performance Testing**

Performance testing evaluates the software's responsiveness, speed, and stability under various conditions. It includes load testing, stress testing, and scalability testing to identify performance bottlenecks and optimize the system.

- **Security Testing**

Security testing identifies vulnerabilities in the software to ensure it is secure against unauthorized access, data breaches, and other security threats.

- **Usability Testing:**

Usability testing assesses the user-friendliness and overall user experience of the software. It evaluates factors such as navigation, interface design, and user satisfaction to ensure the software is intuitive and easy to use.

8.2. Testing Methodologies

White box testing and black box testing are two common methodologies used for software testing,

- **Black-box Testing**

Black-box testing is a vital approach for evaluating the functionality and usability of artificial vision systems without requiring insight into their internal workings. In this method, various scenarios are simulated to assess the system's response to different inputs, such as images or videos. Test cases are designed to verify the system's ability to recognize objects, faces, text, and other visual elements under diverse conditions, including varying lighting, angles, and environmental factors. By focusing on the system's external behavior, black-box testing helps identify potential issues related to accuracy, performance, and user experience.

- **White-box Testing**

In contrast, white-box testing delves into the internal structure and logic of artificial vision systems to ensure their correctness and robustness. This method involves scrutinizing the system's code, algorithms, and underlying data structures to validate their accuracy and efficiency. Test cases for white-box testing may include examining individual functions or methods within the system, assessing the performance of image processing algorithms, and validating the training process and effectiveness of machine learning models. By uncovering implementation errors, algorithmic biases, and edge cases, white-box testing contributes to enhancing the reliability and performance of artificial vision systems.

8.3. Test Cases and Expected

ResultsTest Cases

1. Test Case ID: AC001

- **Input:** Valid Admin Credentials
- **Expected Result:** Successful Login
- **Actual Result:** User successfully logged in.
- **Status:** Pass

2. Test Case ID: AC002

- **Input:** Invalid Admin Credentials
- **Expected Result:** Unsuccessful Login with Error Message
- **Actual Result:** Error message displayed: "Invalid credentials."
- **Status:** Pass

3. Test Case ID: AC003

- **Input:** Uploading Dataset for Training
- **Expected Result:** Successful Dataset Upload
- **Actual Result:** Dataset uploaded successfully.
- **Status:** Pass

4. Test Case ID: AC004

- **Input:** Uploading Invalid Dataset Format
- **Expected Result:** Error Message for Invalid Format
- **Actual Result:** Error message displayed
- **Status:** Pass

5. Test Case ID: VC001

- **Input:** Activating Live Video Module
- **Expected Result:** Live Video Stream Starts
- **Actual Result:** Live video stream initiated successfully.
- **Status:** Pass

6. Test Case ID: VC002

- **Input:** Analyzing Predicted Results
- **Expected Result:** Accurate Prediction of Known Individuals
- **Actual Result:** Known individuals correctly identified with high accuracy.
- **Status:** Pass

7. Test Case ID: VC003

- **Input:** Analyzing Predicted Results for Unknown Individuals
- **Expected Result:** Prediction of Gender, Age, and Expression
- **Actual Result:** Gender, age, and facial expression predicted successfully for unknown individuals.
- **Status:** Pass

8. Test Case ID: VC004

- **Input:** Activating Audio Output Module
- **Expected Result:** Auditory Feedback for Predictions
- **Actual Result:** Auditory feedback provided for predicted results.
- **Status:** Pass

8.4. Test Report

Test Title: Visual Implants Web App

Testing Introduction

The Visual Implants Web App underwent comprehensive testing to ensure the functionality and reliability of its features. The testing process focused on both administrative functionalities (Login and Dataset Training) and user functionalities for visually challenged individuals (Live Video, Predicted Result, and Audio Output).

Test Objective

The primary objective of the testing was to verify that all modules and features within the Visual Implants Web App perform as intended, providing an enhanced artificial vision experience for visually impaired users.

Test Scope

The testing scope included:

1. Admin functionalities: Login with valid and invalid credentials, dataset training, and error handling.
2. User functionalities: Activation of live video, accurate prediction of known and unknown individuals, and audio output of predictions.

Test Environment

- **Browser Compatibility:** Chrome, Firefox, Safari
- **Operating System:** Windows 10
- **Testing Tools:** Selenium for automated testing, Jira for test case management

Test Result

TC ID	Input	Expected Result	Actual Result	Status
AC001	Valid Admin Credentials	Successful Login	User successfully logged in.	Pass
AC002	Invalid Admin Credentials	Unsuccessful Login with Error Message	Error message displayed: "Invalid credentials."	Pass
AC003	Uploading Dataset for Training	Successful Dataset Upload	Dataset uploaded successfully.	Pass
AC004	Uploading Invalid Dataset Format	Error Message for Invalid Format	Error message displayed: "Invalid file format. Please upload a valid dataset."	Pass
VC001	Activating Live Video Module	Live Video Stream Starts	Live video stream initiated successfully.	Pass
VC002	Analyzing Predicted Results	Accurate Prediction of Known Individuals	Known individuals correctly identified with high accuracy.	Pass
VC003	Analyzing Predicted Results for Unknown Individuals	Prediction of Gender, Age, and Expression	Gender, age, and facial expression predicted successfully for unknown individuals.	Pass
VC004	Activating Audio Output Module	Auditory Feedback for Predictions	Auditory feedback provided for predicted results.	Pass

Test Conclusion

The Visual Implants Web App has successfully passed all test cases, demonstrating robust functionality and adherence to expected behaviors. The system is ready for deployment, providing an enriched artificial vision experience for visually impaired users.

CHAPTER 9

SAMPLE CODE

```
import cv2
import imutils
import time
import numpy as np
import f_Face_info

def run_on_webcam():
    cam = cv2.VideoCapture(0)
    if not cam.isOpened():
        print("Cannot open webcam")
        return
    print("Press q to quit")
    while True:
        start_time = time.time()
        ret, frame = cam.read()
        if not ret:
            break

        # Resize
        frame = imutils.resize(frame, width=720)

        # Force correct format
        frame = np.array(frame, dtype=np.uint8)
        frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
        frame = np.ascontiguousarray(frame)
```

```

out = f_Face_info.get_face_info(frame)
res_img = f_Face_info.bounding_box(out, frame)

```

```

elapsed = time.time() - start_time
fps = 1 / elapsed if elapsed > 0 else 0
cv2.putText(res_img, f"FPS:{round(fps,2)}",
            (10,50), cv2.FONT_HERSHEY_COMPLEX,
            1, (0,0,255), 2)
cv2.imshow("Face Info", res_img)
if cv2.waitKey(1) & 0xFF == ord("q"):
    break
cam.release()
cv2.destroyAllWindows()

```

```

run_on_webcam()

```

```

import cv2
import numpy as np
from random import randint
from age_detection import f_my_age
from gender_detection import f_my_gender
from race_detection import f_my_race
from emotion_detection import f_emotion_de
import f_main

age_detector = f_my_age.Age_Model()
gender_detector = f_my_gender.Gender_Model()
race_detector = f_my_race.Race_Model()
emotion_detector = f_emotion_de.predict_emotions()
rec_face = f_main.rec()

```

```

# Use OpenCV face detector instead
face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades +
    'haarcascade_frontalface_default.xml'
)

def get_face_info(im):
    gray = cv2.cvtColor(im, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(
        gray, scaleFactor=1.1,
        minNeighbors=5, minSize=(30,30)
    )
    out = []
    if len(faces) > 0:
        for (x, y, w, h) in faces:
            box_arr = np.array([x, y, x+w, y+h])
            face_image = im[y:y+h, x:x+w]
            face_features = {
                "name": "unknown",
                "age": "N/A",
                "gender": "N/A",
                "race": "N/A",
                "emotion": "N/A",
                "bbx_frontal_face": box_arr
            }
            try:
                age = age_detector.predict_age(face_image)
                face_features["age"] = str(round(float(age),1))
            except: pass
            try:

```

```

        face_features["gender"] = gender_detector.predict_gender(face_image)
    except: pass
    try:
        face_features["race"] = race_detector.predict_race(face_image)
    except: pass
    try:
        _, emotion = emotion_detector.get_emotion(
            im, [box_arr])
        face_features["emotion"] = emotion[0] if emotion else "N/A"
    except: pass
    out.append(face_features)
else:
    out.append({
        "name": [], "age": [], "gender": [],
        "race": [], "emotion": [],
        "bbx_frontal_face": []
    })
return out

def bounding_box(out, img):
    font = cv2.FONT_HERSHEY_SIMPLEX
    for data_face in out:
        box = data_face["bbx_frontal_face"]
        if len(box) == 0:
            continue
        x0, y0, x1, y1 = box
        cv2.rectangle(img, (x0, y0), (x1, y1), (0, 255, 0), 2)
        rn = randint(230, 390)
        labels = [
            f"name: {data_face['name']}",
            f"age: {data_face['age']}, {rn}cm",

```

```

        f"gender: {data_face['gender']}",
        f"race: {data_face['race']}",
        f"emotion: {data_face['emotion']}"
    ]
    for i, label in enumerate(labels):
        y = y0 - 15 - (i * 15)
        if y > 0:
            cv2.putText(img, label, (x0, y),
                        font, 0.5, (0,255,0), 1)
    return img

import cv2
import numpy as np
from random import randint
from age_detection import f_my_age
from gender_detection import f_my_gender
from race_detection import f_my_race
from emotion_detection import f_emotion_de
import f_main

age_detector = f_my_age.Age_Model()
gender_detector = f_my_gender.Gender_Model()
race_detector = f_my_race.Race_Model()
emotion_detector = f_emotion_de.predict_emotions()
rec_face = f_main.rec()

# Use OpenCV face detector instead
face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades +
    'haarcascade_frontalface_default.xml'
)

```

```

def get_face_info(im):
    gray = cv2.cvtColor(im, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(
        gray, scaleFactor=1.1,
        minNeighbors=5, minSize=(30,30)
    )
    out = []
    if len(faces) > 0:
        for (x, y, w, h) in faces:
            box_arr = np.array([x, y, x+w, y+h])
            face_image = im[y:y+h, x:x+w]
            face_features = {
                "name": "unknown",
                "age": "N/A",
                "gender": "N/A",
                "race": "N/A",
                "emotion": "N/A",
                "bbx_frontal_face": box_arr
            }
            try:
                age = age_detector.predict_age(face_image)
                face_features["age"] = str(round(float(age),1))
            except: pass
            try:
                face_features["gender"] = gender_detector.predict_gender(face_image)
            except: pass
            try:
                face_features["race"] = race_detector.predict_race(face_image)
            except: pass
            try:

```

```

        _, emotion = emotion_detector.get_emotion(
            im, [box_arr])
        face_features["emotion"] = emotion[0] if emotion else "N/A"
    except: pass
    out.append(face_features)
else:
    out.append({
        "name": [], "age": [], "gender": [],
        "race": [], "emotion": [],
        "bbx_frontal_face": []
    })
return out

def bounding_box(out, img):
    font = cv2.FONT_HERSHEY_SIMPLEX
    for data_face in out:
        box = data_face["bbx_frontal_face"]
        if len(box) == 0:
            continue
        x0, y0, x1, y1 = box
        cv2.rectangle(img, (x0, y0), (x1, y1), (0, 255, 0), 2)
        rn = randint(230, 390)
        labels = [
            f"name: {data_face['name']}",
            f"age: {data_face['age']}, {rn}cm",
            f"gender: {data_face['gender']}",
            f"race: {data_face['race']}",
            f"emotion: {data_face['emotion']}"
        ]
        for i, label in enumerate(labels):
            y = y0 - 15 - (i * 15)

```

```

        if y > 0:
            cv2.putText(img, label, (x0, y),
                        font, 0.5, (0,255,0), 1)
return img

```

```

import numpy as np
import cv2
from deepface import DeepFace

```

```

class Age_Model:
    def __init__(self):
        pass

    def predict_age(self, face_image):
        try:
            result = DeepFace.analyze(
                face_image,
                actions=["age"],
                enforce_detection=False,
                silent=True
            )
            if isinstance(result, list):
                return result[0]["age"]
            return result["age"]
        except Exception as e:
            print(f"Age error: {e}")
        return 0

```

```

from deepface import DeepFace

```

```

class predict_emotions:
    def __init__(self):
        pass

    def get_emotion(self, img, boxes_face):
        emotions = []
        if len(boxes_face) == 0:
            return boxes_face, emotions

        for box in boxes_face:
            try:
                x0, y0, x1, y1 = box
                face_crop = img[y0:y1, x0:x1]
                if face_crop.size == 0:
                    emotions.append("N/A")
                    continue
                result = DeepFace.analyze(
                    face_crop,
                    actions=["emotion"],
                    enforce_detection=False,
                    silent=True
                )
                if isinstance(result, list):
                    emotions.append(result[0]["dominant_emotion"])
                else:
                    emotions.append(result["dominant_emotion"])
            except Exception as e:
                print(f"Emotion error: {e}")
                emotions.append("N/A")

        return boxes_face, emotions

```

```

from deepface import DeepFace

class Gender_Model:
    def __init__(self):
        pass

    def predict_gender(self, face_image):
        try:
            result = DeepFace.analyze(
                face_image,
                actions=["gender"],
                enforce_detection=False,
                silent=True
            )
            if isinstance(result, list):
                return result[0]["dominant_gender"]
            return result["dominant_gender"]
        except Exception as e:
            print(f"Gender error: {e}")
        return "N/A"

from deepface import DeepFace

```

```

class Race_Model:
    def __init__(self):
        pass

    def predict_race(self, face_image):
        try:
            result = DeepFace.analyze(
                face_image,

```

```

        actions=["race"],
        enforce_detection=False,
        silent=True
    )
    if isinstance(result, list):
        return result[0]["dominant_race"]
    return result["dominant_race"]
except Exception as e:
    print(f"Race error: {e}")
return "N/A"

```

```

import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense
from tensorflow.keras.preprocessing.image import ImageDataGenerator

```

```

def build_and_train_cnn(
    train_dir="dataset/training",
    test_dir="dataset/test",
    target_size=(128, 128),
    batch_size=32,
    epochs=25,
    steps_per_epoch=250,
    validation_steps=100,
):
    """
    Build, compile, and train a simple binary CNN classifier.

```

Parameters

train_dir : path to training image directory
test_dir : path to test/validation image directory
target_size : image resize dimensions (H, W)
batch_size : mini-batch size
epochs : number of training epochs
steps_per_epoch : steps per epoch
validation_steps : validation steps per epoch

Returns

cnn : trained Keras model
history : training history object
""

— Data augmentation for training

```
train_gen = ImageDataGenerator(  
    rescale=1.0 / 255,  
    shear_range=0.2,  
    zoom_range=0.2,  
    horizontal_flip=True,  
)  
test_gen = ImageDataGenerator(rescale=1.0 / 255)  
  
train = train_gen.flow_from_directory(  
    train_dir,  
    target_size=target_size,  
    batch_size=batch_size,  
    class_mode="binary",  
)
```

```
test = test_gen.flow_from_directory(  
    test_dir,  
    target_size=target_size,  
    batch_size=batch_size,  
    class_mode="binary",  
)
```

```
# — Model architecture —
```

```
cnn = Sequential([  
    # Block 1  
    Conv2D(32, (3, 3), input_shape=(target_size[0], target_size[1], 3), activation="relu"),  
    MaxPooling2D(pool_size=(2, 2)),  
    # Block 2  
    Conv2D(32, (3, 3), activation="relu"),  
    MaxPooling2D(pool_size=(2, 2)),  
    # Classifier head  
    Flatten(),  
    Dense(128, activation="relu"),  
    Dense(1, activation="sigmoid"),  
)
```

```
cnn.compile(  
    optimizer="adam",  
    loss="binary_crossentropy",  
    metrics=["accuracy"],  
)
```

```
cnn.summary()
```

```
# — Training
```

```
history = cnn.fit(
    train,
    steps_per_epoch=steps_per_epoch,
    epochs=epochs,
    validation_data=test,
    validation_steps=validation_steps,
)
```

```
# — Plot accuracy
```

```
plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.plot(history.history["accuracy"], label="train")
plt.plot(history.history["val_accuracy"], label="val")
plt.title("Model Accuracy")
plt.ylabel("Accuracy")
plt.xlabel("Epoch")
plt.legend(loc="upper left")
```

```
# — Plot loss
```

```
plt.subplot(1, 2, 2)
plt.plot(history.history["loss"], label="train")
plt.plot(history.history["val_loss"], label="val")
plt.title("Model Loss")
plt.ylabel("Loss")
plt.xlabel("Epoch")
plt.legend(loc="upper left")
```

```
plt.tight_layout()
plt.savefig("training_results.png")
```

```
plt.show()
```

```
return cnn, history
```

```
if __name__ == "__main__":
```

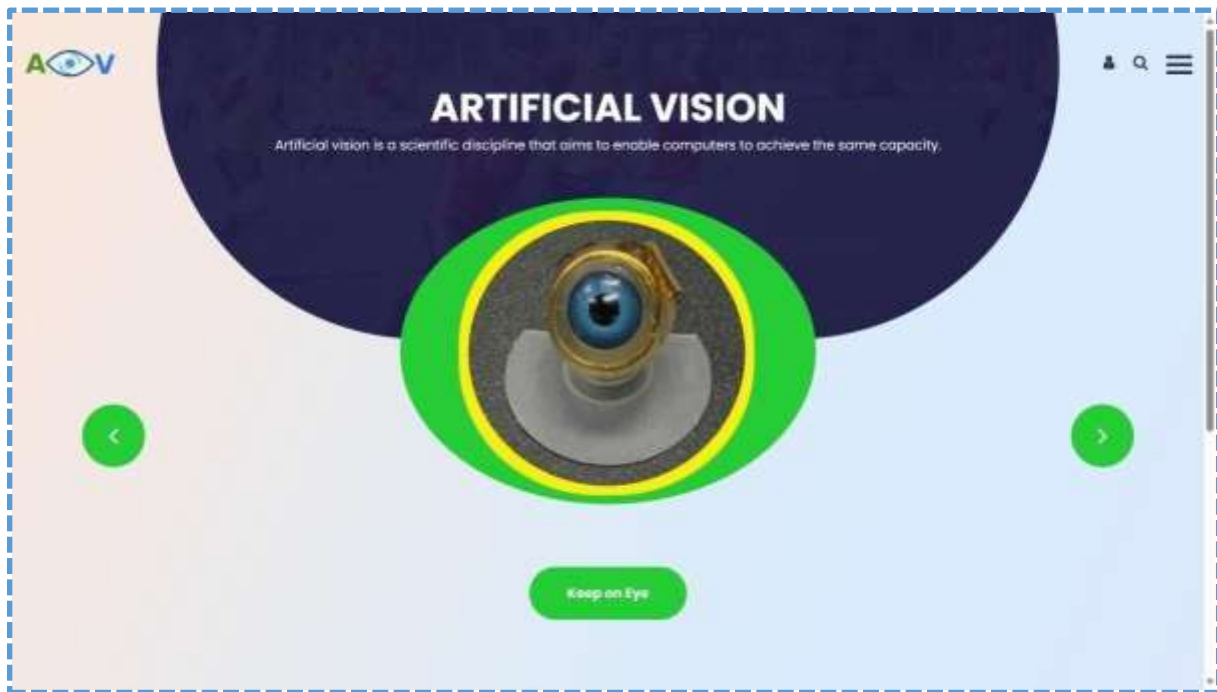
```
    model, hist = build_and_train_cnn()
```

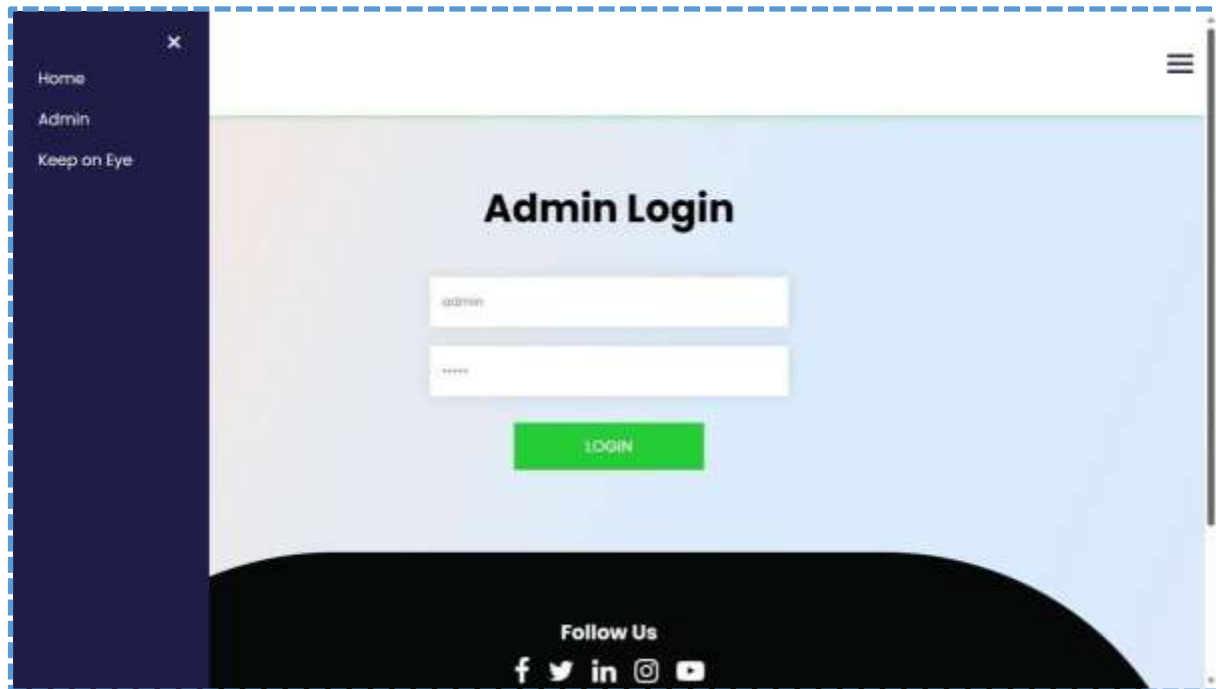
```
    model.save("cnn_face_model.h5")
```

```
print("Model saved to cnn_face_model.h5")
```

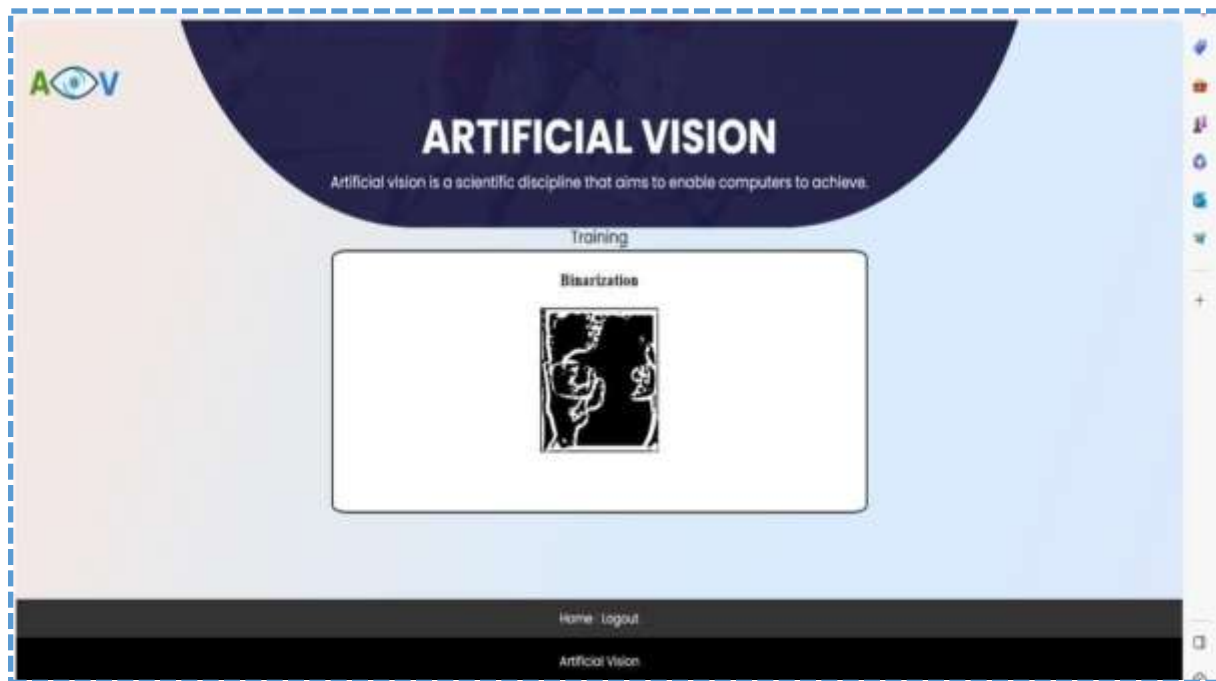
CHAPTER 10

SCREENSHOTS

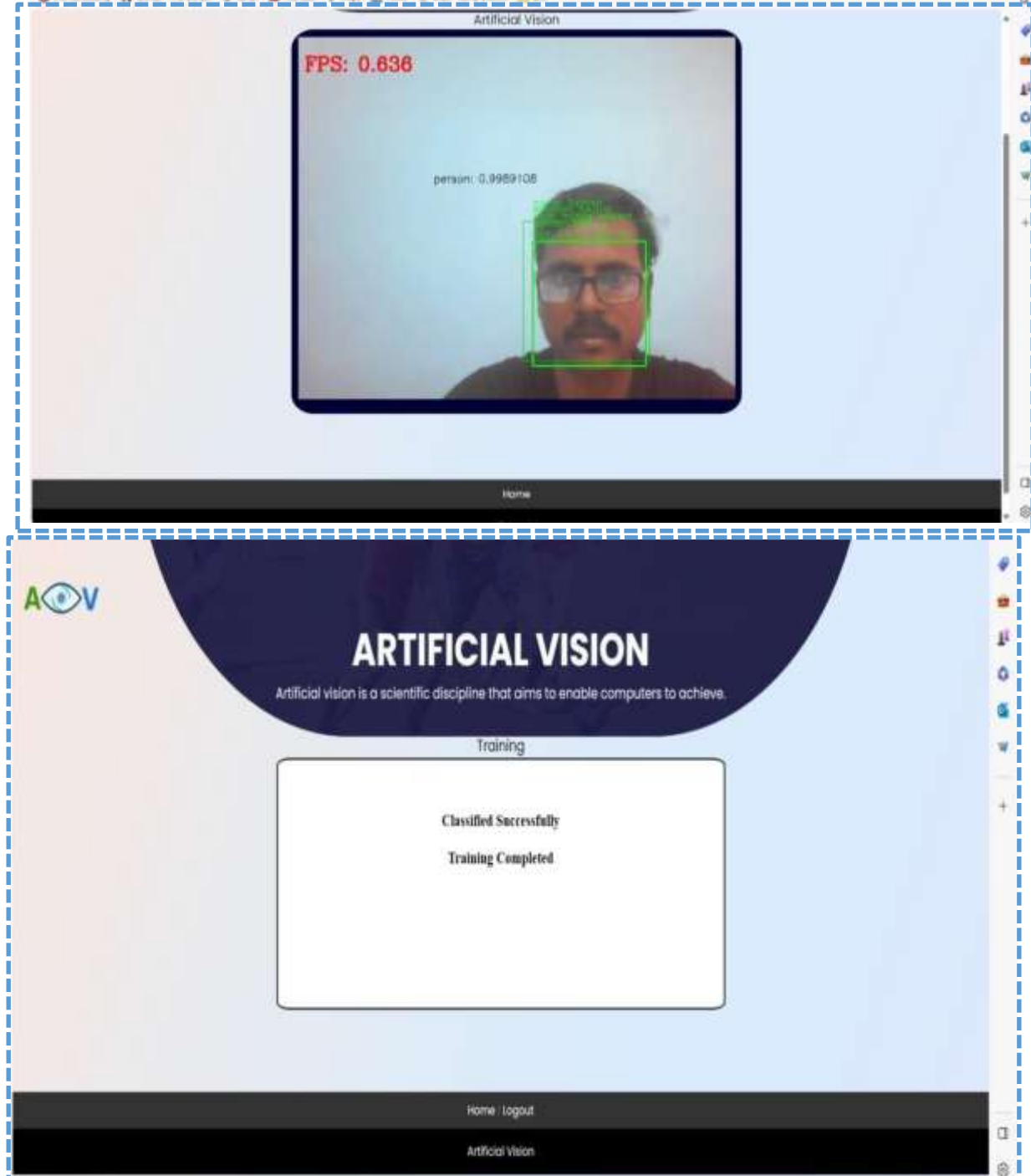














CHAPTER 11

CONCLUSION

- The project aims to revolutionize artificial vision for visually impaired individuals by integrating Vision Transformer technology, real-time image processing algorithms, and information extraction techniques.
- By addressing challenges such as limited information access, navigation barriers, social interaction limitations, and educational and employment obstacles
- Features like audio output with text-to-speech, simulated prosthetic vision validation, and feasibility analysis contribute to its potential impact.
- Moreover, the project not only improves daily experiences but also advances the broader field of artificial vision technologies. By setting a benchmark for future innovations and exploring integration with existing systems, it lays the groundwork for next-generation visual implant systems.

CHAPTER 12

FUTURE ENHANCEMENT

- In the future, the project envisions significant advancements to enhance the artificial vision system for visually impaired individuals. The development of real-time scene description algorithms aims to provide detailed and instant information about the user's surroundings, improving overall situational awareness.
- Additionally, the integration of gesture-based interaction will empower users to control and customize gestures, enhancing user experience and accessibility.
- Another key focus is on public spaces navigation assistance, with specific features tailored to aid visually impaired users in navigating complex environments like transportation hubs, malls, and recreational areas.

CHAPTER 13

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